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Artificial Neural Network Inducement for Enhancement of Cloud Computing Security

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***Abstract*—Cloud computing is a spirited topic in this epoch where IOT introducing the modern lifestyle. These embrace cost-effectiveness, time savings and the actual arrangement of computing properties. Conversely, in Cloud Computing privacy and security is a big issue where there are many kinds of thread and attack are working rapidly. In the context, the best cryptographic method AES algorithm and AI-based method ANN have chosen to work for the enhancement of the security where cloud computing demands the highest priority. In this case, this paper is proposing the highly advanced two-step security layer for cloud computing. Primarily, AES provides for the first layer security of the first stratum. In the AES algorithm, the execution is dependent on the key size of the algorithm when the number of rounds to be accomplished. A MATLAB code is developed for plaintext encryption and cipher text decryption. Experiments are conducted to measure execution times. ANN is a method of computation which is biologically stimulated. These only look like the parallel computation generated which are the basics of human learning by the biological neural network. Iris and Finger recognition is implemented using MATLAB through ANN. The paper demonstrated the great proficiency of the proposed system for the user purpose.**

Keywords—Cloud Computing Security, Artificial Neural Network, Advanced Encryption Standard, Iris Recognition, Fingerprint Recognition

I. Introduction

The world is more conscious about enriching the security issue. Cloud computing is the most burning topic to ensure the security of it. To do that many algorithms applied to the platform. Many of them are secured but not fast, many of them fast but not secured. So the research is continued till now on this security system. The proposed method of the paper is to increase both fast and secure algorithm.

In 1997, the National Institute of Standards and Technology decided to organize a competition to find out the successor of DES algorithm. Where the Rijndael algorithm (designed by Vincent Rijmen and Joan Daemen in 1977) had won the competition in April 2000 and would, later on, become the AES. The cryptographic community was assured the algorithm has been thoroughly tested and screened by the NIST to still be secure even 2030 [1].

AES (Advanced Encryption Standard) is the advanced, fast, latest and most secure system at the current time. The Fast Encryption system is too far to meet with the encryption time of conventional AES algorithm. Like as, most valuable research topic become the high-performance computing capability of graphics processing unit [2]. The security system used on the first layer of the proposed method is to encryption the traditional login system with a password. AES operates on a 4 × 4 column-major order matrix of bytes, designated by the state, although some versions of Rijndael have a larger block size and have extra columns in the state. Most AES calculations are done in a particular finite field.

The second layer is based on Iris recognition and Fingerprint based on biometric tools. Iris recognition is a preset method of biometric identification. It uses mathematical pattern-recognition techniques on images of one or both of the Irises of an entity's eyes. where, patterns are unique, stable, and can be understood from some distance. A biometric system is designed for automatic authentication over the special characteristic option of the human body. The most reliable and proficient Biometric system is Iris recognition. There are some parts; spherical Iris and pupil region, occluding eyelids and eyelashes and reflections of Iris are extremely automatic segmentation where the Hough transform and is capable to localize. [3]. So this will help the system to make more efficient to the best security system. In the method firstly the system will capture the Iris to register for the future login. In the second verification of login, there will the first option for verifying the identity.

On the other hand, if anyone is unable to do Iris scan, there is another option to verify is Biometric Fingerprint scanning. The Biometric authentication of the user provides some facility like as fast, precise, easy to use, trustworthy and economical where¬ this is conventional knowledge-based and token-based methods. [4].

II. Related Works

In this proposed work we introduce the security enhancement of user side access to cloud computing. There are well known and developed algorithms AES for the initial step to encrypt and decrypt the password and another one is Artificial Neural Network (ANN) using for implementing the Fingerprint and Irish recognition.

In [5], proposed a method based on neural networks for detection and classification of underwater mines using sonar images is proposed. The paper shows the success of the classification rate using neural networks.

Likewise, in [6] shows the neural network system with supervised and unsupervised learning is used, so that the presentations among these are associated, shown the same spectral data in the input.

In Y. LeCun et al [7] they recognize handwrite using convolution neural network, which provides a good result. And another one is similarly P. Vincent [8], they proposed a

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method to showing the advantage and stacked de-noising auto encoders model in handwriting recognition. Although there are a variety of deep structure models [9], they all gain very good results in many fields, such as visual [10], natural language processing [11]and signal processing and so on. So we believe that the method based on depth neural network can also achieve good results for the fingerprint classification.

Bodade and Talbar [12] proposed a method for recognizing different kinds of iris images, spoofs and eyes on the video. This paper shows 3 categories to spoof detection are the reflectance method, the pupil dynamics method, and the frequency spectrum method. They achieve a mishmash of the methods is needed and shows some outcomes of iris images recognition.

Daugman approaches cosmetic contact lenses recognition [13]. This paper shows the working procedure for lenses where dot-matrix type technology is used for manufacturing. However, there are different types of approaches for manufacture the contact lenses and dot-matrix type lenses also can be failed as if they printed resolution of the iris sensor more than the camera resolution

III. METHODOLOGY

Here, the paper will design to express the real-life strong method to secure cloud computing technology. This system will narrate in several sections includes encryption, the process of registration and verification with the provided database. This system is shown in some flowcharts and diagram.

A. *Initial Encryption and Decryption:*

The basic level of the providing of cloud computing security is designed with Advanced Encryption Security (AES). Which one is the existing conventional method for encryption and decryption work on the first layer of our proposed method? Password and Username will be stored in the database when the user will provide.

B. *User Registration procedure:*

The user will get in this process after completing the initial step. This layer is designed with 2 option where the biometric system Iris recognition and Fingerprint recognition will be worked for registration. The explanation and working procedure are briefly narrating with each step include Figure-1:

Step-1: The Encrypted password will be stored in the database. Successfully completing this, there will be 2 option to choose as a process of registration.

Option-1: The Neural Network algorithm will be applied to recognize the Iris. Here, our system will take some different position, a movement image of Iris. Which will be saved in the Database of the root user.

Option-2: The Neural Network algorithm will be applied for this option for the user registration. Where this process will take some sample of Finger to execute this method. This sample also saves in the database.

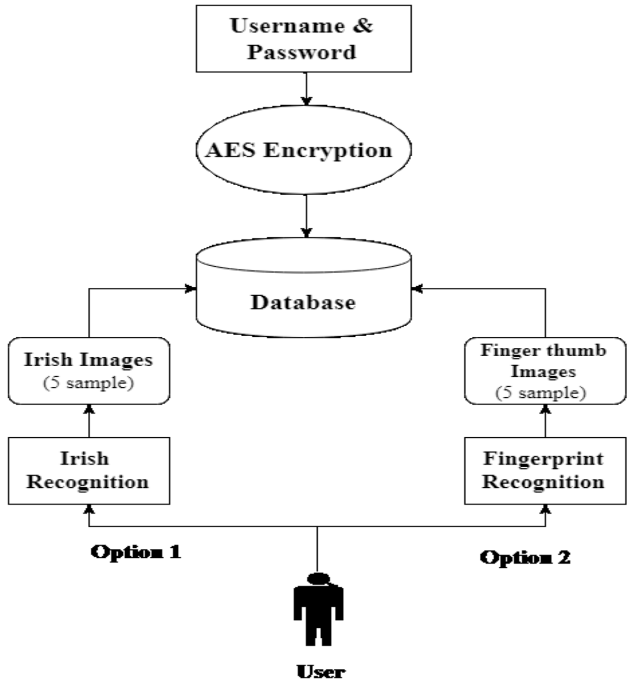

Figure-1: Registration Cycle

C. *Data Matching:*

Matching is the very crucial parts of the proposed model where the design requirements to verify with stored data from the database with the input data by comparing. This procedure will show in the figure and it will explain several steps.

Step-1: The username and password will be the first step of the Data-matching process where AES will be executed for encryption and decryption. The user will give his password and username which will be cross match through the Database. When the particular password verified with the database, the user gets access from the method and step-2 begins. But, if not verified with the database, then "Login Error" will be shown.

Step-2: When successfully complete the authentication, the user will able to get through the Biometric verification with one option out of two.

Option-1: This portion works on user Iris recognition where the user's Iris picture will be captured using the Digital camera. here, there are a number of pictures will be reserved. The Neural Network is applied as a pattern recognition with the database. If the Iris matched enough, then the user will get the confirmation to get access to "Login". On the other hand, if the algorithm gives a big number of false position rate on Iris. It will not verify which will indicate "Login Error".

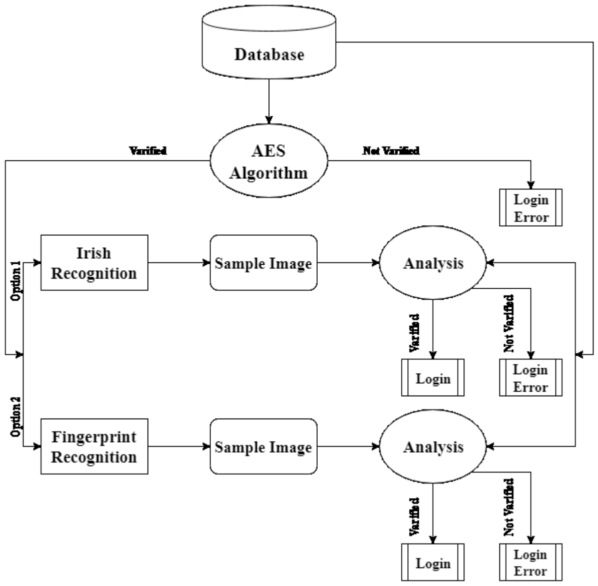

Figure-2: Data Matching process.

Option-2: This step is another Biometric step where the Neural Network algorithm will apply for the pattern recognition on the Fingerprint of the user. If the Iris matched enough, then the user will get the confirmation to get access to "Login". On the other hand, if the algorithm gives a big number of false position rate on Iris. It will not verify which will indicate "Login Error".

IV. Implementation

This part will represent the full proposed method where we designed a system where it will find out the user by using three cited the most secure option. These methods will be shown for AES, Neural Network for Iris recognition and Fingerprint recognition. Figure-1,2,3 defines how the implementation process is verified in the actual object.

A. Advance Security System (AES)

AES is a Symmetric Algorithm, which uses encryption and decryption a data or text by a similar key for both sender and receiver. This is the most proficient algorithm aimed at software application. [14]

AES algorithm is a Block cipher and works on a fixed-length set of bits. It proceeds an input block of 128 bits and yields a matching output block of the similar size. This system calls for a second input, which called a secret key. Moreover, AES has three different key sizes: 128, 192 and 256 bits. This time, Algorithm works on around function for 4 different byte-oriented layers through the cipher and inverse cipher:

1) Byte substitution using a substitution table (Sbox)

2) Shifting rows of the State array by different offsets

3) Mixing the data within each column of the State array

4) Adding a Round Key to the State. AES algorithm uses this round function

for ten rounds, but the first and last round of the AES algorithm is differing from other rounds. That is depicted in figure 1 below. This template was designed for two affiliation

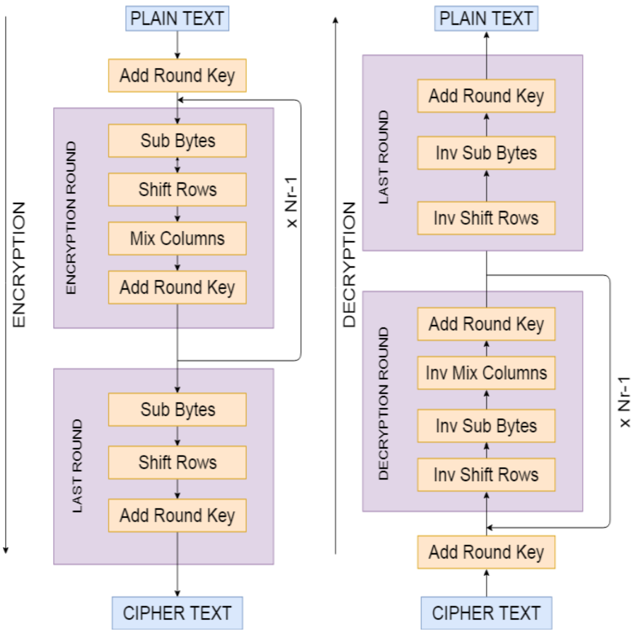

Figure-3: Advanced Encryption Standard (AES) working flowchart.

B. Neural Network

The artificial neural network works on a neuron which is taken for input and changes it's some of the internal state names as Activation based on the input, besides create the output dependent on input and Activation. Basically, a directed weighted graph consists of the network which customs by linking that input of additional neurons to the neuron's output. The weights can be improved by a system known as learning that computes the activation and directed by a learning rule. [15]

ANN is capable of learning and it needs to be trained. This system runs on several strategies:

- Supervised Learning
- Unsupervised Learning
- Reinforcement Learning

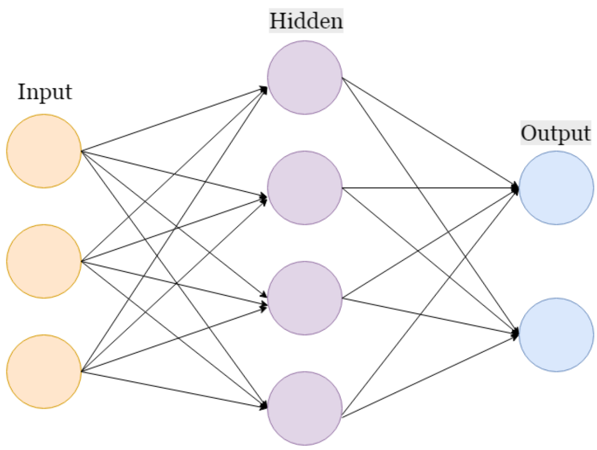

Figure-4: Working process of Artificial Neural Network.

Backpropagation: The backpropagation's weight updates operated by stochastic gradient descent over the equation:

$$\omega_{ij}(t+1) = \omega_{ij}(t) + \eta \frac{\partial C}{\partial \omega_{ij}} + \xi(t)$$

where,

η = the learning rate

C = the cost (loss) function

$\xi(t)$ = stochastic term.

The cost function dependent on influence and the activation. Influence is generated by supervised, reinforcement, etc. Such as, if executing a function on a multi-class arrangement of the problem, cost function and common selection of activation are defining consistently, cross entropy function and softmax function. The softmax is shown as

$$\rho_j = \frac{\exp(x_j)}{\sum_j d_j \log(x_k)}$$

where ρ_j is the class probability (output of the unit j and signify the total input to units j and k of the same level correspondingly. [16]

C. Finger Recognition

The fingerprint is the most common and exclusive method in Biometric options which is transnationally known as a legal method to identify a person. The reproductions of the minute ridge which called as Dermal of the Finger. Ridges and Valleys are distinctive and unchangeable. [17] Finger ridge ending as a ridge point is established ends sharply. Ridge splits are mention as ridge point which is connected with the branch ridges. The minutiae location, attributes and time like a direction can be shown in a Fingerprint. There are 40 to 100 minutiae in a standard quality fingerprint sample, on the other hand, dozens of minutiae cannot identify the pattern of the fingerprint. [18]

Implementation of the Fingerprint verification system is broken down into four distinct stages as follows:

- Image acquisition.
- Edge detection.
- Comparison of images.
- Decision making.

D. Iris Recognition

Iris biometric technology was established by John Daugman [19]. Iris biometric is a texture of the Iris can be used as an underlying idea where the code is generated based on the image condition. Iris authentication becomes a progressively spirited research topic in the new era [20]. The exhausting of different types of contact lenses present defies for Iris biometric systems [21]. "Cosmetic contact lenses" used for the change Iris behavior of others in any way. Challenges to identify the image of eye uncovered with cosmetic contact lenses with the image of the similar eye covered with cosmetic contact lens outcome in an extremely wrong nonmatch rate [21]. That similarly imaginable, an attacker trying to imitate a specific user possibly will have cosmetic contact lenses completed to achieve the victim person's Iris grain. This is called "spoofing" of the Iris biometric system. The cornea is the outermost (anterior) layer of the eye. It is normally clear; this system provides some portions of the Iris sample. The aqueous humor is following the layer of the cornea. normally clear; this system affords some percentages of the Iris sample. The following layer is Iris (and pupil). Where the quality of the Iris is imaged over the cornea and aqueous humor. [22]

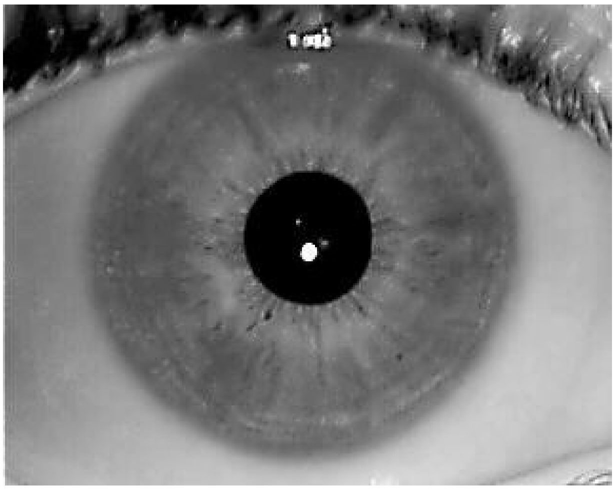

Figure-5: The Structural position of Iris.

Implementation of the Fingerprint verification system is broken down into four distinct stages as follows:

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- Decision making.

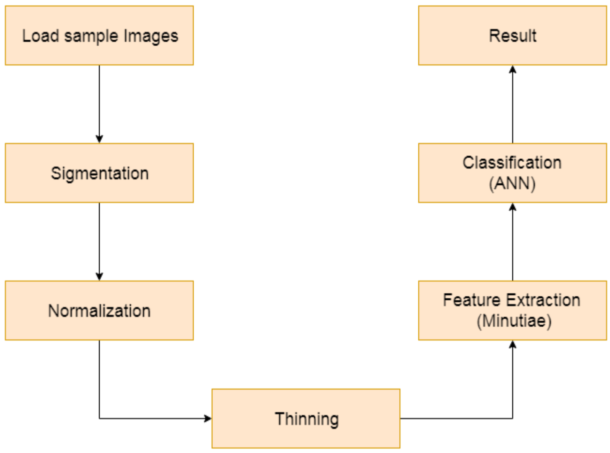

Figure-6: The Working procedure of Finger and Iris Recognition by using ANN.

In MATLAB the iteration is conducted by our own created tow dataset of separately Fingerprints and Irises. Here, we established the iteration is conducted by the sample image matrix binary values. Figure-7 depicts the whole procedure of Fingerprint and Iris recognition mechanism using Artificial Neural Network with some stepwise operations.

The Experimental dataset for the experiment is created our won using camera and fingerprint scanner. Our working station configuration is 7th gen core i3-7020u processor (3m cache,2.30 GHz); 4GB DDR4 RAM, 1TB HDD with windows 10 setup. In the dataset, our data samples are in

.png format images. We collected different peoples' data use to making the data set.

Following steps (1-4) establish how MATLAB works to recognize the user's iris and finger samples from the database. For the successful experiment, we have developed two personal datasets to effectively iris and fingerprint in MATLAB. These steps are provided for fingerprint recognitions where will be similar steps for iris recognition too.

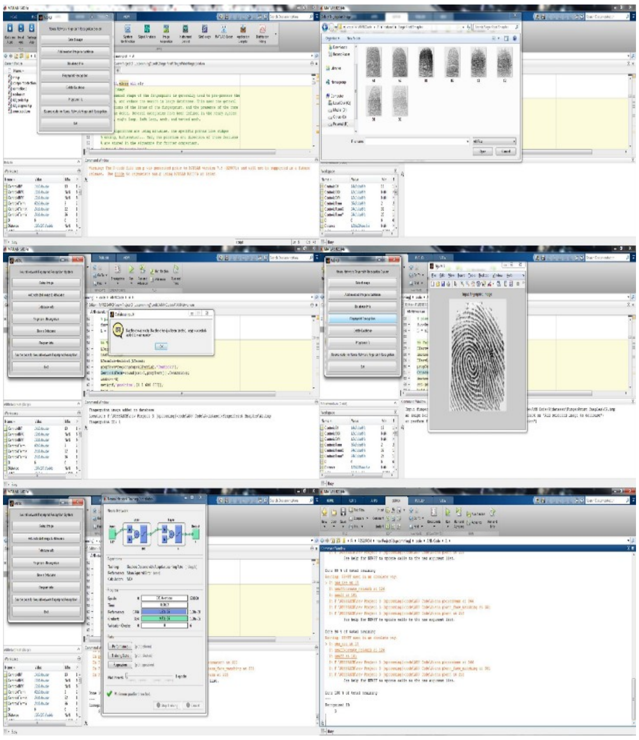

Figure-7: ANN implementation in MATLAB.

Step-1: Executing and administration the codes of ANN to add some samples for the experiment.

Step-2: All samples are in Bitmap (. bmp) image format for our farther steps. Extract and upload system of samples are followed by the database. Here, we provide four different samples for the dataset according to A1, A2; B1, B2; C1, C2; D1, D2.

Samples Name	Dataset Name
A1	1
B2	2
C1	3
D1	4
A2	5

Step-3: Comparing two samples, namely sample 1.bmp with sample 5.bmp, Then the result confirmations that mentioned samples are matched with Sample 1. So the resolution designates the matching of experiment samples.

Step-4: Whenever the samples didn't match, results show the unsuccessful matching results.

V. Experimental Results

In the starting, the proposed method is to implement the AES algorithm. AES has just two operations, encryption, and decryption of the password of each user. After that, the Neural Network toolbox in MATLAB is implemented, where Artificial Neural Network is created for Iris Recognition and biometric Fingerprint. A customized dataset with randomized value is used in case of an experiment and got a great output at the end of the experiment. The experiment is based on time and efficiency issue.

TABLE-1: 1st Layer Encryption, Decryption Analysis

	Iteration	Encryption Cycle (ms)	Decryption Cycle (ms)	Size (KB)
AES Algorithm (256bit)	01	160	110	363
	02	170	140	613
	03	220	180	944

Here, TABLE-1 is representing the result of encryption and decryption cycle of an AES 256 algorithm. Algorithms run for three iterations. For the encryption of AES algorithm gradually takes 160ms, 170ms and finally 220ms for according to 163kb, 313kb and 944kb where, decryption cycle gradually takes time 100ms, 140ms, and 180ms. Where this system provides feedback as a very speedy method.

TABLE-2: 2nd Layer Comparison Analysis

Steps	Iris Recognition (ms)	Finger Recognition (ms)	Size (KB)
Artificial Neural Network Encryption	1134	1465	2654
Artificial Neural Network Decryption	334	255	2654

In the 2nd layer, table-2 is representing the encryption and decryption cycle of the proposed method. Using AI, encryption takes 1134ms for 2654kb for Iris and 1465ms for fingerprint method. On the other hand, the decryption cycle takes 334ms for Iris and 255ms for Fingerprint matching.

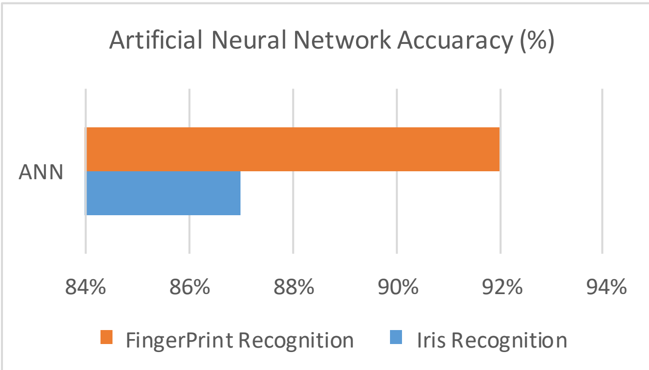

Figure-8: Accuracy Performance of ANN

Here, in the figure-8 shows the accuracy of the recognition is great. Thus the accuracy is showing that 87% accurate for Iris Recognition and for 92% for Fingerprint in biometric technology.

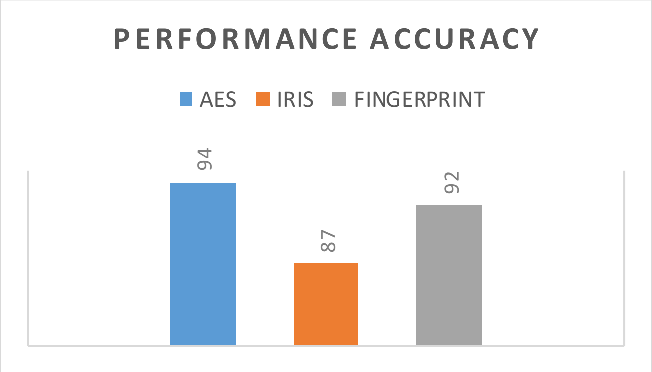

Figure-9: Accuracy Comparison Graph

According to Figure-9; Finally, the performance graph is showing the difference in the final result of the accuracy percentage of the result with the previous method comparison. And the graph surely shows that the proposed method is better working and reliable.

VI. Conclusions

In this paper, the importance of Cloud Computing security has been revealed strongly. Through the proposal method, the first layer AES algorithm provides high performance and accuracy. In the other hand, the second layer; Iris recognition or Fingerprint recognition through Artificial Neural Network will be responsible for the best accuracy of secure access for the user. This proposed procedure can accomplish the secure access of the most uses cloud computing interface for the client users.

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References

[1] Saini, G. & Singh, P. (2014). Audio Steganography by LSB method and enhanced security with AES. International Journal of Advanced Research in Computer Science & Technology, 2(2), 255-258. Retrieved March 2015.

[2] AES Encryption Algorithm Based on the High-Performance Computing of GPU. Electronic ISBN: 978-1-4244-5727-4, Fei Shao, Zinan Chang, Yi Zhang.

[3] Human Iris recognition for biometric identification. ISBN: 978-1-4244-1550-2. Computer and information technology, 2007. iccit 2007. 10th international conference. Paul Padma Polash, Md. Maruf Monwar.

[4] A Study of Biometric Approach Using Fingerprint Recognition Ravi Subban and Dattatreya P. Mankame. Lecture Notes on Software Engineering, Vol. 1, No. 2, May 2013.

[5] S. Geethalakshmi, P. Subashini, and S. Ramya, “A Study on Detection and Classification of Underwater Mines Using Neural Networks,” Int. J. Soft Comput. Eng. IJSCE, vol. 1, no. 5, pp. 150–157, 2006.

[6] B. Howell, S. Wood, and S. Koksal, “Passive sonar Recognition and analysis using hubrid neural networks,” IEEE Oceans, vol. 4, pp. 1917–1924, 2003.

[7] LeCun Y, Bottou L, Bengio Y, et al. Gradient-based learning applied to document recognition[J]. Proceedings of the IEEE, 1998, 86(11): 2278-2324.

[8] Vincent P, Larochelle H, Bengio Y, et al. Extracting and composing robust features with denoising autoencoders [C]//Proceedings of the 25th international conference on Machine learning. ACM, 2008: 1096-1103.

[9] Bengio Y. Learning deep architectures for AI[J]. Foundations and trends R in Machine Learning, 2009, 2(1): 1-127.

[10] Yang J, Yu K, Gong Y, et al. Linear spatial pyramid matching using sparse coding for image classification[C]//Computer Vision and Pattern Recognition, 2009. CVPR 2009. IEEE Conference on. IEEE, 2009: 1794- 1801.

[11] Collobert R, Weston J. A unified architecture for natural language processing: Deep neural networks with multitask learning[C]//Proceedings of the 25th international conference on Machine learning. ACM, 2008: 160-167.

[12] R. Bodade and S. Talbar, “Fake Iris Detection: A Holistic Approach” International Journal of Computer Applications 19 (2), April 2011.

[13] J. Daugman, “Demodulation by Complex-Valued Wavelets for Stochastic Pattern Recognition”, International Journal of Wavelets, Multiresolution and Information processing 1 (1), 2003.

[14] Karthigaikumar, P. & Rasheed, S. (2011). Simulation of image encryption using AES algorithm. Computational Science - New Dimensions and Perspectives, 166. Retrieved March 2015.

[15] Zell, Andreas (1994). "chapter 5.2". Simulation Neuronaler Netze {Simulation of Neural Networks} (in German) (1st ed.). Addison-Wesley. ISBN 3-89319-554-8.

[16] Hinton, G.; Deng, L.; Yu, D.; Dahl, G. E.; Mohamed, A. r; Jaitly, N.; Senior, A.; Vanhoucke, V.; Nguyen, P. (November 2012). "Deep Neural Networks for Acoustic Modeling in Speech Recognition: The Shared Views of Four Research Groups". IEEE Signal Processing Magazine. 29 (6): 82–97. Bibcode:2012ISPM...29...82H. doi:10.1109/msp.2012.2205597. ISSN 1053-5888.

[17] A. K.Jain, S. Pankanti, and R. Bolle, Biometrics: Personal Identification in Networked Society[M]. Boston: Kluwer Academic Publishers, 1999.

[18] R. Thai, “Fingerprint Image Enhancement and Minutiae Extraction”, 2003

[19] J. G. Daugman, “Biometric Personal Identification System Based on Iris Analysis”, US Patent # 5,291,560, March 1, 1994], [J. Daugman, “How Iris Recognition Works”, IEEE Transactions on Circuits and Systems for Video Technology 14 (1), Jan. 2004, 21-30.

[20] M.J. Burge and K.W. Bowyer, Eds., Handbook of Iris Recognition, Springer, 2013. ISBN 978-1-4471-4401-4.

[21] S. E. Baker, A. Hentz, K. W. Bowyer, and P. J. Flynn,“Degradation of Iris Recognition Performance Due to Non-Cosmetic Prescription Contact Lenses,” Computer Vision and Image Understanding 114 (9), Sept. 2010,1030-1044.

[22] k.Hughes,KW.Bowyer “Detection of Contact-Lens-Based Iris Biometric Spoofs Using Stereo Imaging” System Sciences (HICSS), 2013 46th Hawaii International Conference on, 18 March. 2013, 13397903.